



ECEN 314—Signals and Systems

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Module 23

Application: Radar

Learning Objectives



- Understand how radar systems use time delay estimation to measure target range.
- Understand how radar systems use the Doppler effect to measure target velocity.
- Understand how FMCW radar systems simultaneously estimate the range and velocity of a target.

Radar Basics



Many cars these days are built with a radar system.

A signal is transmitted, propagates to the other vehicle, reflects and then returns to the original source.

By comparing the original and reflected signals we can determine:

- ✓ distance
- ✓ velocity



- We can determine the distance by measuring the round trip delay.
- We can determine the relative velocity by measuring the Doppler (frequency) shift.
- Q: What sort of signal should we transmit that would make it easiest to extract the information we are interested in?



Continuous Wave (CW) Radar



Suppose we transmit a single frequency (sinusoid) over some fairly long interval of time:

$$s(t) = A \cdot \cos(\omega_o t + \theta), \quad 0 \leq t \leq T_s.$$

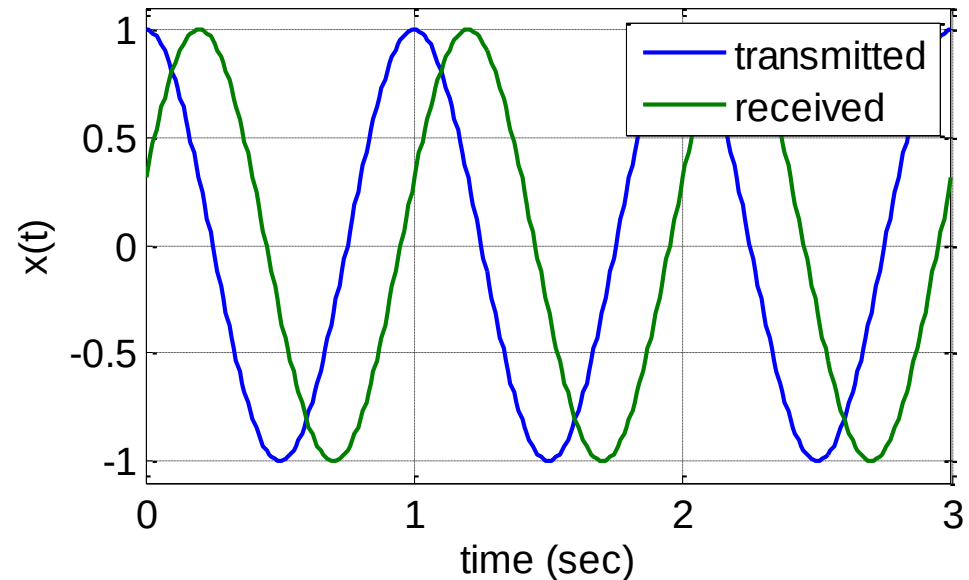
If the transmitter and the target are separated by a distance with no relative motion, the received signal will be

$$r(t) = B \cdot \cos(\omega_o(t - \Delta t) + \theta),$$

with the round trip propagation time given by

$$\Delta t = \frac{2 \cdot d_o}{c},$$

where c is the propagation speed ($\approx$ speed of light for EM waves).



The distance can be measured by estimating the phase shift between the two signals.

$$\Delta\theta = \frac{2 \cdot d_o \cdot \omega_o}{c}$$

To avoid ambiguity, we could measure distances up to

$$d_{max} = \frac{c}{2f_o}$$

Continuous Wave (CW) Radar



If the target is in motion relative to the transmitter with a velocity, v , then the distance to the target is time varying

$$d(t) = d_o + vt.$$

The received signal is then

$$\begin{aligned} r(t) &= B \cdot \cos\left(\omega_o\left(t - \frac{2(d_o + vt)}{c}\right) + \theta\right) \\ &= B \cdot \cos\left(\left(\omega_o\left(1 - \frac{2v}{c}\right)\right)t + \theta - \frac{2d_o\omega_o}{c}\right) \end{aligned}$$

In addition to the phase shift, there is now a frequency shift (Doppler shift) of

$$\Delta\omega = \frac{2v\omega_o}{c}$$

Using typical numbers:

$$\begin{aligned} f_o &= 4\text{GHz} \\ d_o &= 30\text{m} (\approx 100\text{ft}) \\ v &= 75\text{mph} (33.3\text{m/s}) \end{aligned}$$

Maximum range estimate:

$$d_{max} = \frac{c}{2f_o} = \frac{3 \cdot 10^8\text{m/s}}{2 \cdot 4 \cdot 10^9\text{Hz}} = 3.75\text{cm}$$

Doppler Shift:

$$\begin{aligned} \Delta f &= \frac{2vf_o}{c} = \frac{2 \cdot 33.3\text{m/s} \cdot 4 \cdot 10^9\text{Hz}}{3 \cdot 10^8\text{m/s}} \\ &= 888\text{Hz} \end{aligned}$$

✓ Using a sinusoid would be good for measuring velocity, but not distance.

Pulse Radar



Next, suppose we sent a short duration pulse,

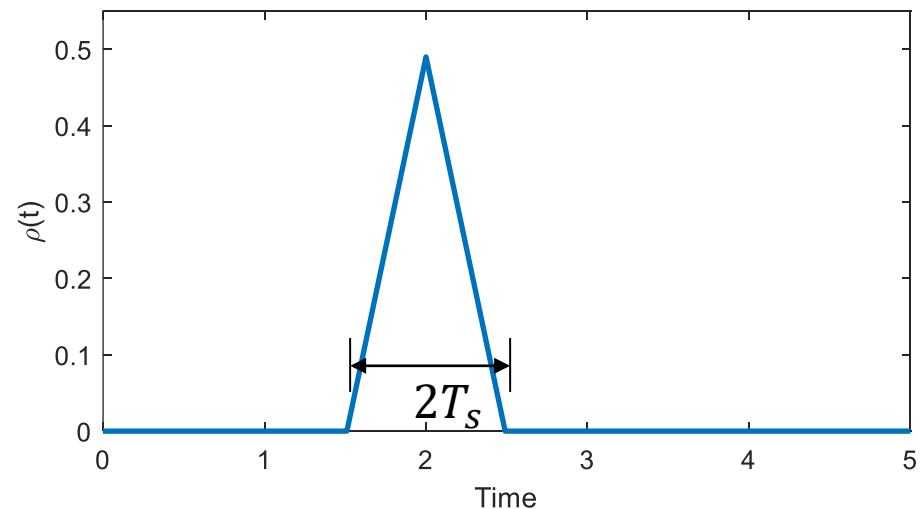
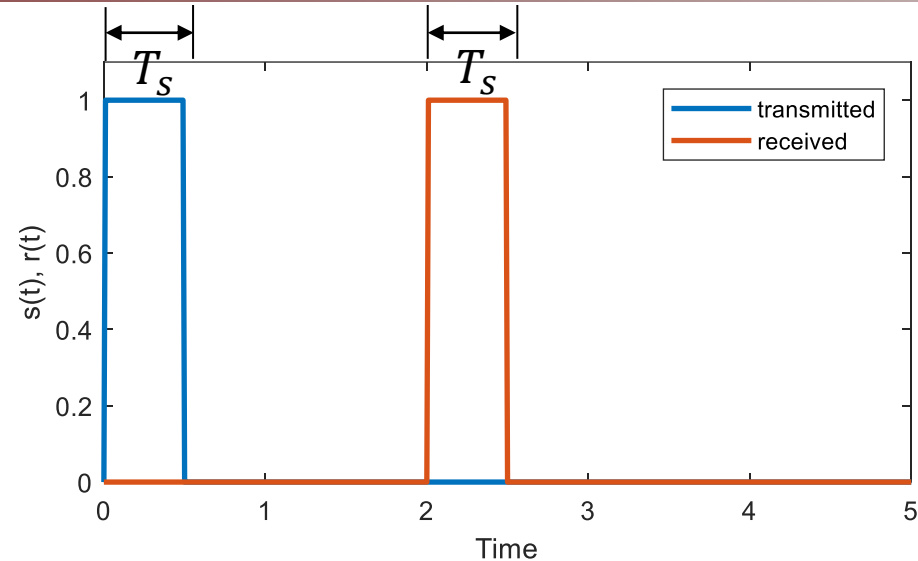
$$s(t) = A \cdot \text{rect}\left(\frac{t - T_s/2}{T_s}\right)$$

With no relative motion, the received signal would be

$$r(t) = A \cdot \text{rect}\left(\frac{t - \Delta t - T_s/2}{T_s}\right)$$

A common method to estimate relative time delay between two signals is to use an operation called *correlation* (very similar to convolution)

$$\rho(t) = \int_{-\infty}^{\infty} r(u)s(u - t)du$$



Pulse Radar



If the target is in motion relative to the transmitter, the received signal is then

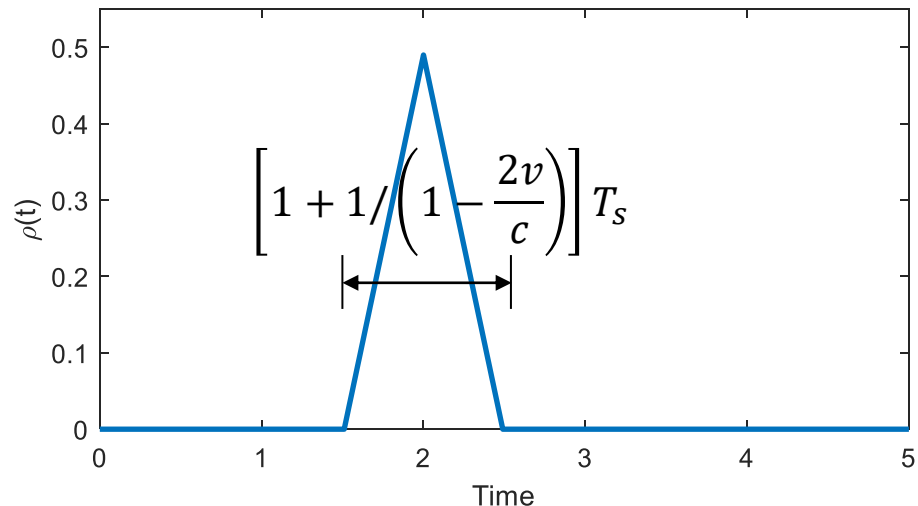
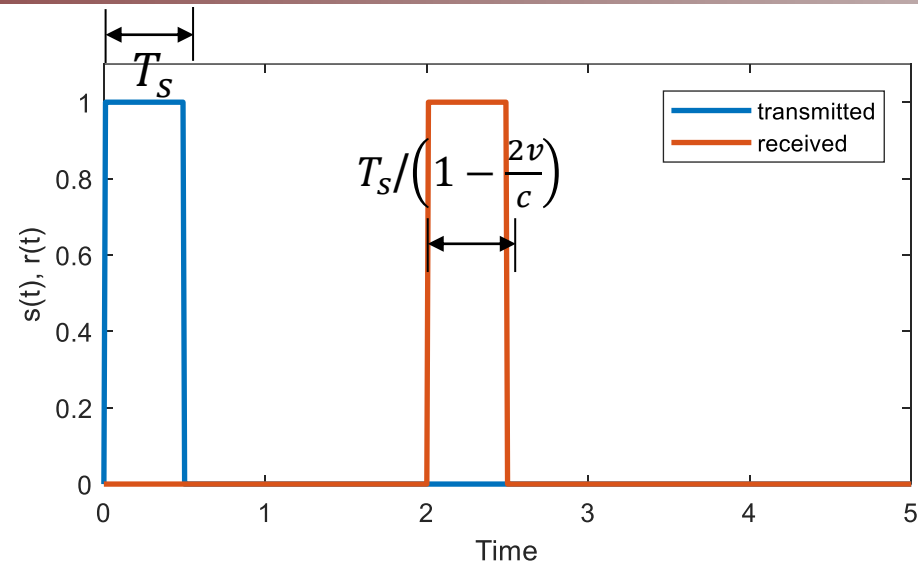
$$r(t) = B \cdot \text{rect} \left(\frac{t - \frac{2(d_o + vt)}{c} - T_s/2}{T_s} \right)$$

$$= B \cdot \text{rect} \left(\frac{t \left(1 - 2\frac{v}{c} \right) - \frac{2d_o}{c} - T_s/2}{T_s} \right)$$

In addition to the time delay, the width of the received pulse is expanded or compressed by a factor of $1 - 2\frac{v}{c}$.

Since $v \ll c$, the change in the width of the pulses would be very difficult to measure.

- ✓ A narrow pulse is good for measuring distance, but not velocity.

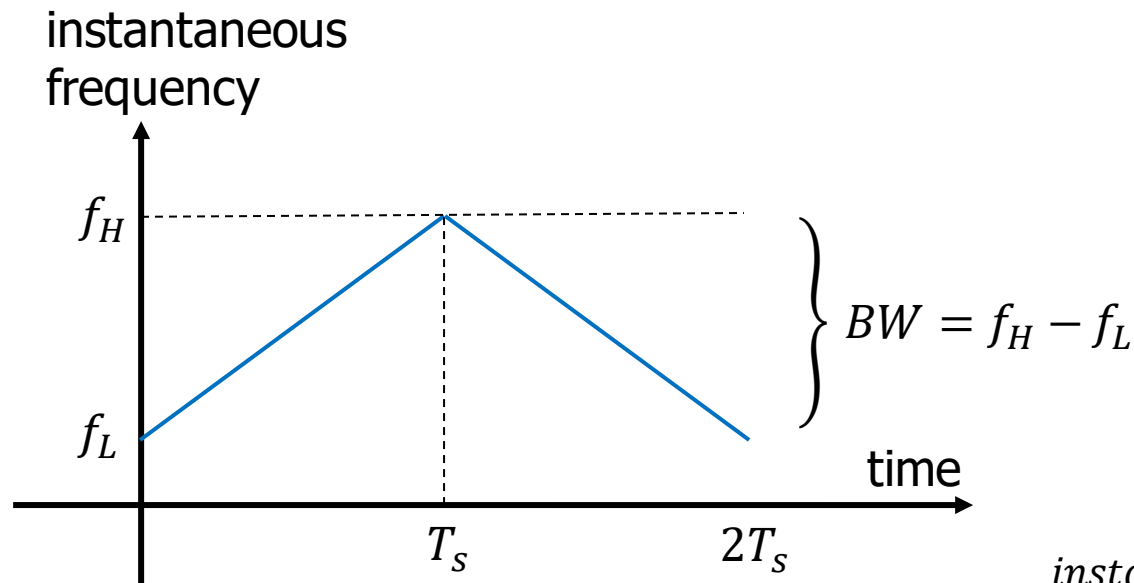


FMCW (Chirp) Radar



It is common to use what is called a frequency modulated continuous wave (FMCW) signal, also known as a *chirp* signal.

A chirp is an FM signal where the frequency varies linearly with time.



$$s(t) = A \cdot \cos\left(\pi \left(\frac{f_H - f_L}{T_s}\right) t^2 + 2\pi f_L t\right), \quad 0 \leq t < T_s$$

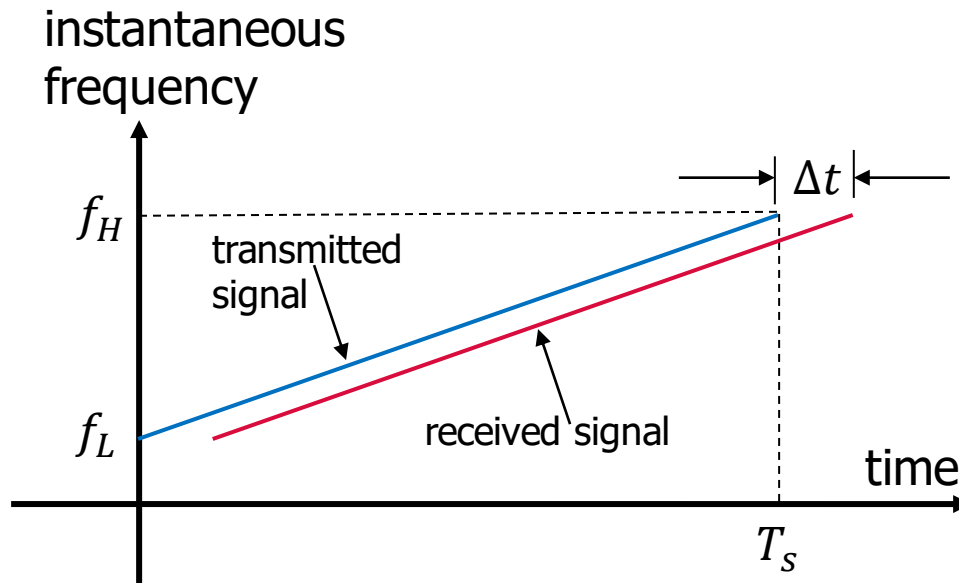
The instantaneous frequency of the chirp signal advances linearly from some starting value, f_L , to some final value, f_H , over a time duration of T_s seconds, and then descends back over the next T_s seconds (and then repeats the process).

$$\begin{aligned} &\text{instantaneous frequency} \\ &= \begin{cases} \frac{f_H - f_L}{T_s} t + f_L, & 0 \leq t < T_s, \\ -\frac{f_H - f_L}{T_s} (t - T_s) + f_H, & T_s \leq t < 2T_s. \end{cases} \end{aligned}$$

FMCW Radar – Measuring Distance



If there is no relative motion between transmitter and object, the received signal will be a time delayed (and maybe attenuated) version of the transmitted signal.



The transmitted and received signals are separated by a constant frequency difference.

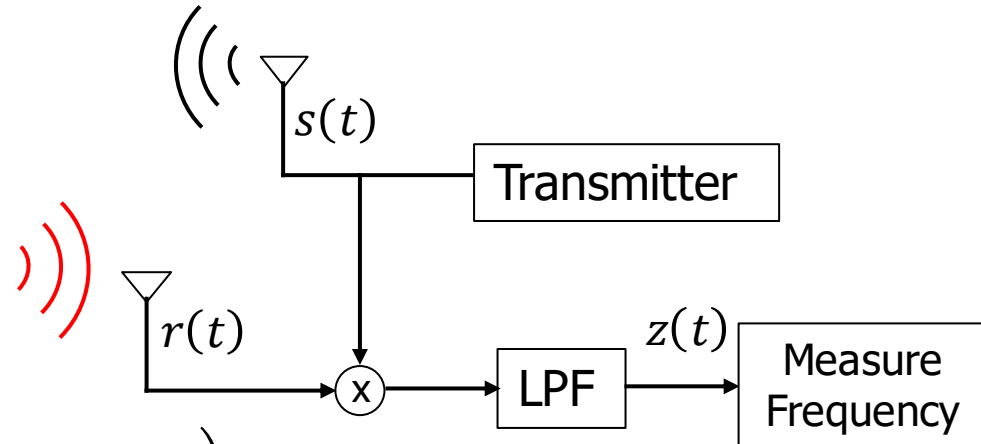
How do we measure the frequency difference between two signals?

$$s(t) = 2 \cdot \cos\left(\pi \left(\frac{f_H - f_L}{T_s}\right) t^2 + f_L t\right) \quad r(t) = B \cdot \cos\left(\pi \left(\frac{f_H - f_L}{T_s}\right) (t - \Delta t)^2 + f_L (t - \Delta t)\right)$$

FMCW Radar – Measuring Distance



We can use the simple architecture shown to detect the frequency difference between the transmitted and received signal.



$$s(t) = 2 \cdot \cos\left(\pi\left(\frac{f_H - f_L}{T_s}\right)t^2 + 2\pi f_L t\right)$$

$$r(t) = B \cdot \cos\left(\pi\left(\frac{f_H - f_L}{T_s}\right)(t - \Delta t)^2 + 2\pi f_L(t - \Delta t)\right)$$

$$s(t)r(t) = B \cos\left(\pi\left(\frac{f_H - f_L}{T_s}\right)[(t - \Delta t)^2 + t^2] + 2\pi f_L(2t - \Delta t)\right)$$

$$+ B \cos\left(\pi\left(\frac{f_H - f_L}{T_s}\right)[2\Delta t \cdot t - \Delta t^2] + 2\pi f_L \cdot \Delta t\right)$$

$$z(t) = B \cos\left(2\pi\left(\frac{\Delta t}{T_s}\right)(f_H - f_L)t + \theta\right)$$

The LPF blocks the high frequency term and passes the low frequency term.

The output of the LPF will be at a frequency of

$$f_z = \left(\frac{\Delta t}{T_s}\right)(f_H - f_L)$$

FMCW Radar – Measuring Distance



Now that we have a signal $z(t)$ that is a sinusoid, how do we measure its frequency?

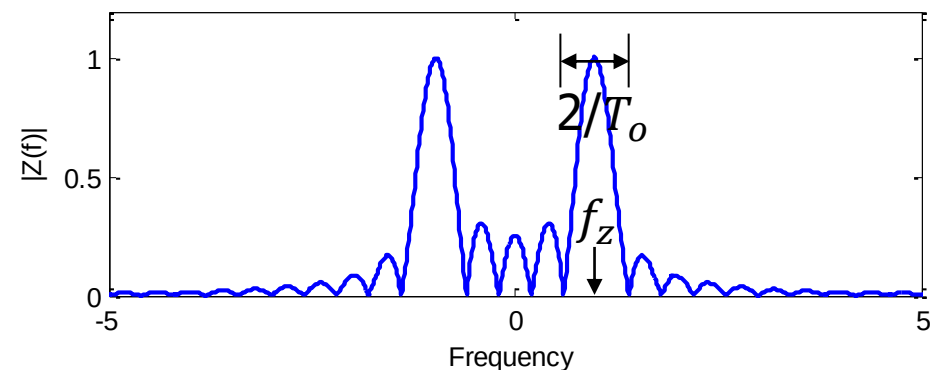
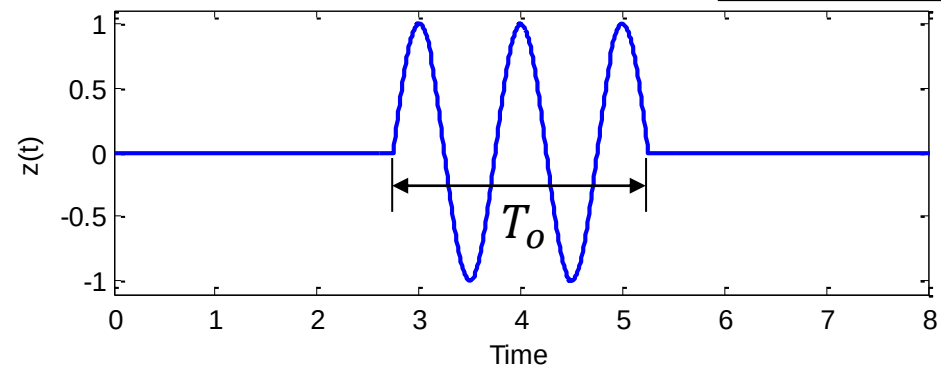
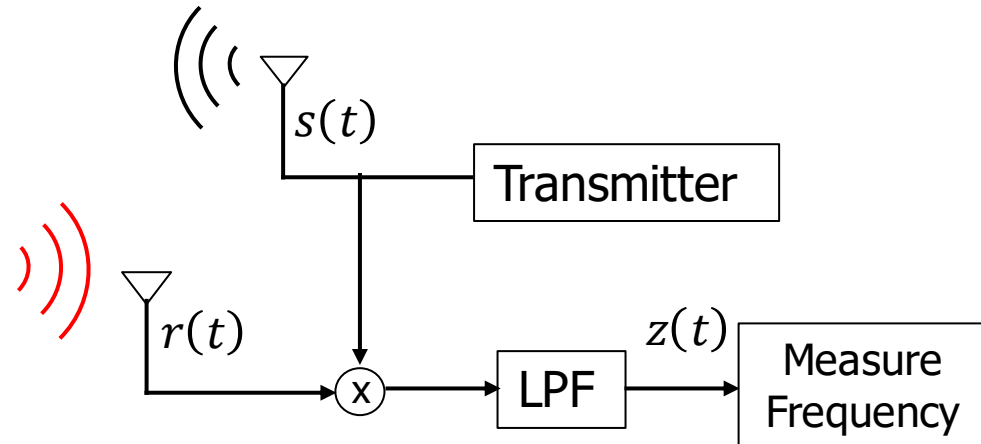
$$z(t) \propto \cos(2\pi f_z t + \theta)$$

$$f_z = \left(\frac{\Delta t}{T_s} \right) (f_H - f_L)$$

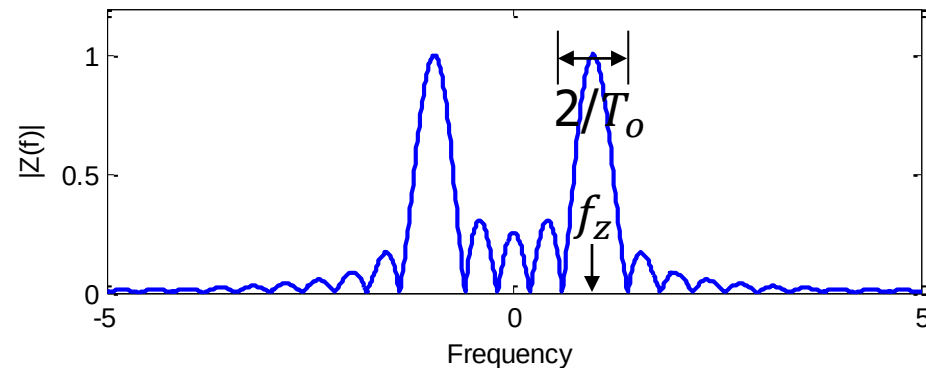
We could observe the signal for some period of time, T_o , and then take it's Fourier Transform which should exhibit a maximum near the frequency, f_z .

Once f_z is estimated, the distance to the object is then given by

$$d = \frac{f_z T_s c}{2(f_H - f_L)}$$



FMCW Radar – Measuring Distance



How long does our observation interval, T_o , need to be?

- ✓ We would like to make our estimate as quickly as possible so that our information is not outdated by the time we get it.
- ✓ If T_o is too small, the FT will not have sharp peaks and it will be difficult to get an accurate estimate of distance.

$$1/T_o < f_z \rightarrow T_o > 1/f_z$$

Some typical numbers:

Object:

Distance = 30m ($\approx 100ft$)

$$\rightarrow \Delta t = \frac{2d}{c} = \frac{2(30m)}{3 \cdot 10^8 m/s} = 0.2 \mu s$$

Velocity = 75mph (33.3 meters/sec)

$\rightarrow$ object will move 3.3 cm every msec.

$$\rightarrow T_o < 100ms$$

FMCW Radar System:

$$f_H - f_L = 100MHz$$

$$T_s = 1msec \rightarrow T_o < 1ms$$

$$\rightarrow f_z = \left(\frac{\Delta t}{T_s} \right) (f_H - f_L)$$

$$= \frac{(0.2 \mu s)(100MHz)}{1msec} = 20kHz$$

$$T_o > \frac{1}{f_z} = 50 \mu s$$

FMCW Radar – Range Resolution



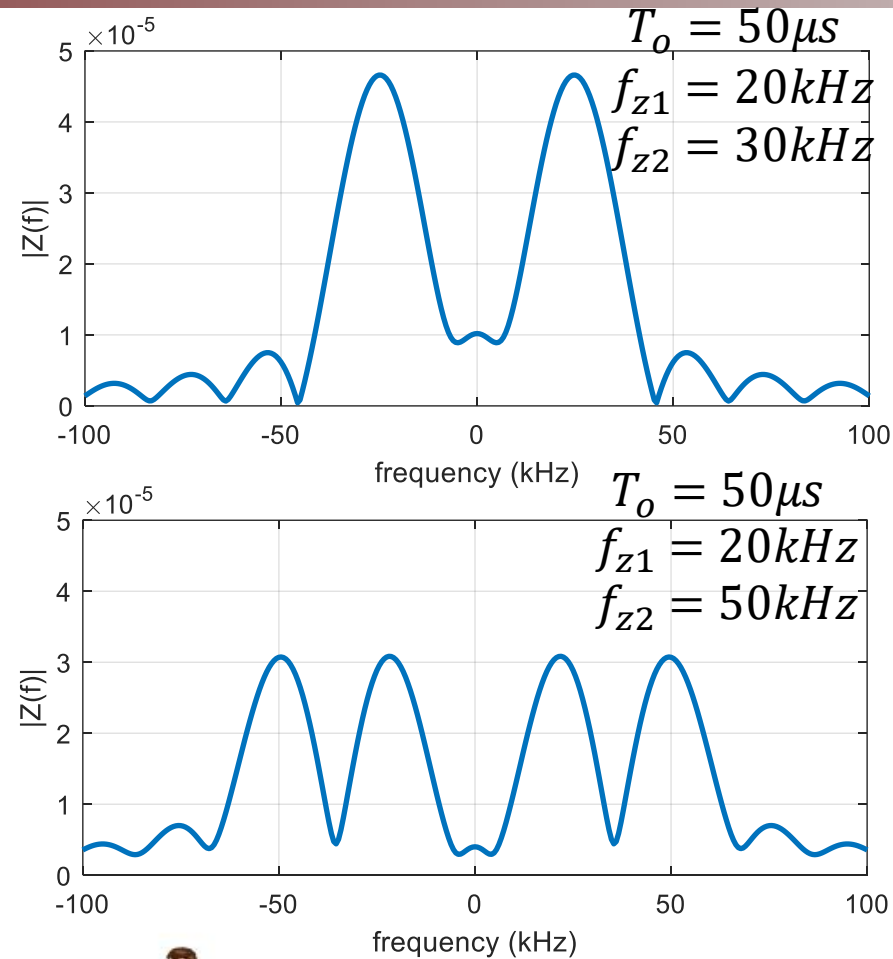
Now suppose there are two targets at different distances. Our received signal will look like

$$r(t) = B_1 s(t - \Delta t_1) + B_2 s(t - \Delta t_2)$$

After we mix the received signal with the transmitted signal, we will now see the sum of two sinusoids

$$z(t) = B_1 \cos(\omega_{z1} t + \theta_1) + B_2 \cos(\omega_{z2} t + \theta_2)$$

If the targets are at similar distance, they look like a single target. What can we change about our system to help resolve close targets?

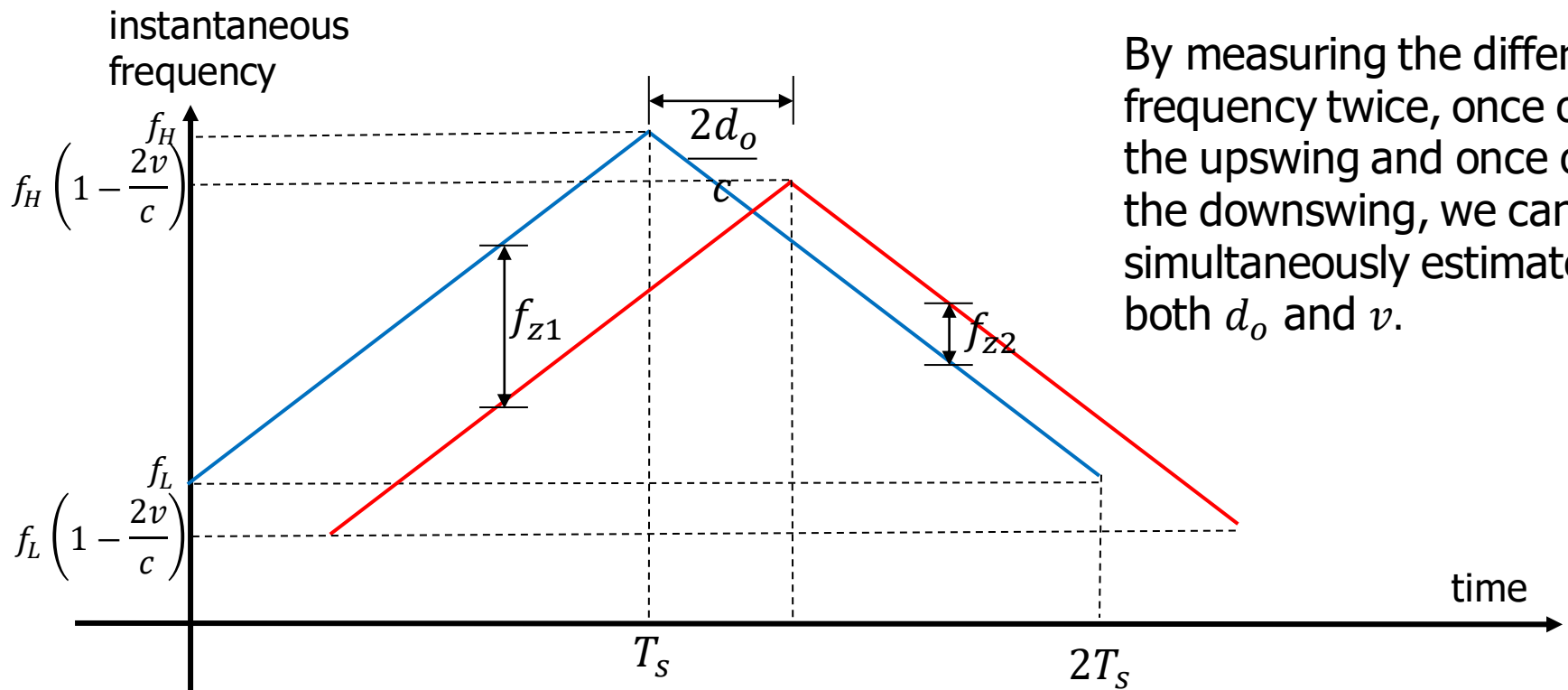


FMCW Radar – Range/Velocity Measurement



Next, suppose our target is in motion (relative to the transmitter). The instantaneous frequency of the reflected signal will differ from the transmitted signal in two ways:

- 1) Due to time delay from round trip propagation (shift right by $\frac{2d_o}{c}$),
- 2) Due to Doppler effect (shift down or up proportional to v).



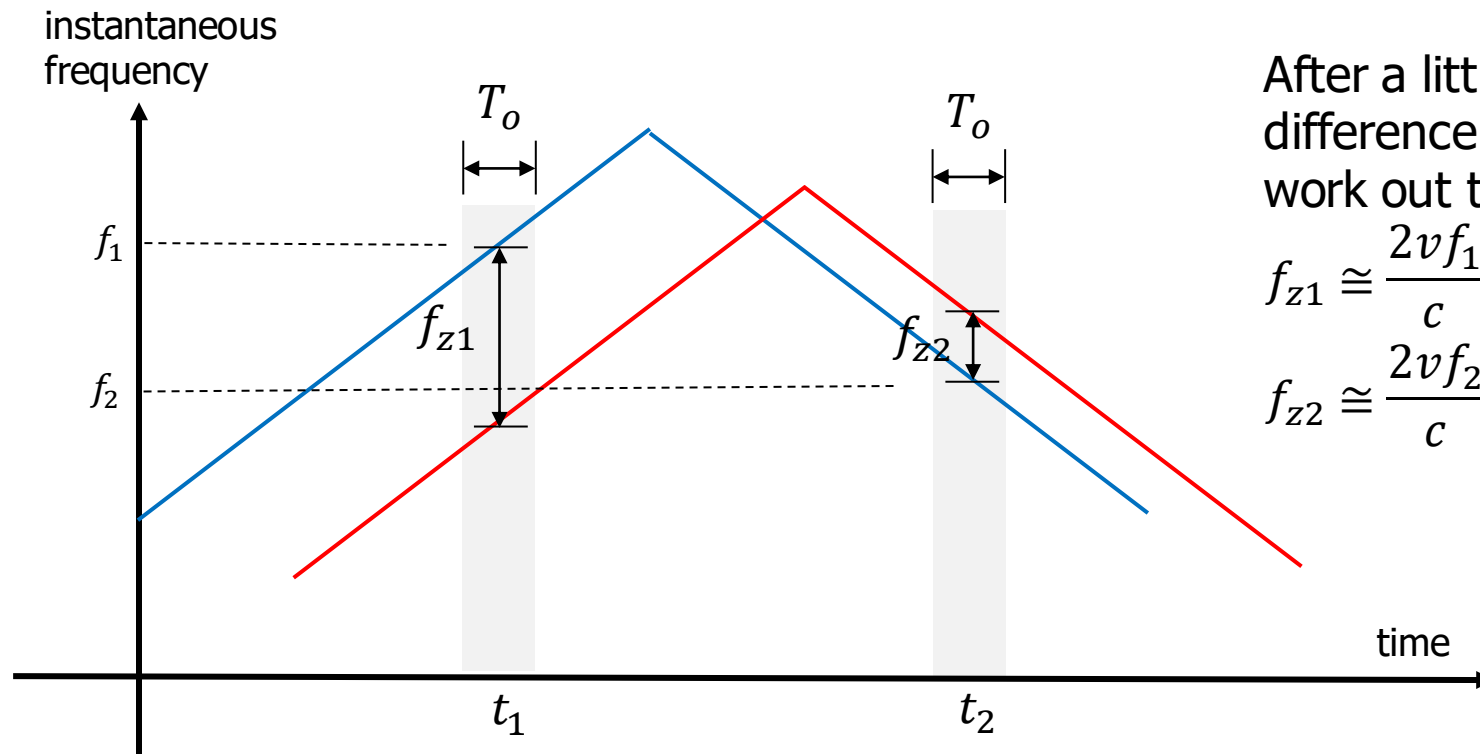
By measuring the difference frequency twice, once on the upswing and once on the downswing, we can simultaneously estimate both d_o and v .

FMCW Radar – Range/Velocity Measurement



Suppose we observe the difference frequencies during some observation interval around some time $t = t_1$ during the upswing and some time $t = t_2$ during the downswing. Then define:

- ✓ f_1 = transmitted frequency at time $t = t_1$,
- ✓ f_2 = transmitted frequency at time $t = t_2$,
- ✓ f_{z1} = observed difference frequency at time $t = t_1$,
- ✓ f_{z2} = observed difference frequency at time $t = t_2$.



After a little algebra, the difference frequencies work out to be:

$$f_{z1} \cong \frac{2vf_1}{c} + \frac{2d_o}{c} \cdot \frac{f_H - f_L}{T_s}$$

$$f_{z2} \cong \frac{2vf_2}{c} - \frac{2d_o}{c} \cdot \frac{f_H - f_L}{T_s}$$

FMCW Radar – Range/Velocity Measurement



The system of equations can easily be solved for the distance and velocity of the target.

$$\begin{bmatrix} f_{z1} \\ f_{z2} \end{bmatrix} = \frac{2}{c} \begin{bmatrix} f_1 & \frac{f_H - f_L}{T_s} \\ f_2 & -\frac{f_H - f_L}{T_s} \end{bmatrix} \begin{bmatrix} v \\ d_o \end{bmatrix} \quad \rightarrow \quad \begin{aligned} v &= \frac{c}{2} \cdot \frac{f_{z1} + f_{z2}}{f_1 + f_2} \\ d_o &= \frac{cT_s}{2(f_H - f_L)} \cdot \frac{f_2 f_{z1} - f_1 f_{z2}}{f_1 + f_2} \end{aligned}$$

Example:

System parameters	Measurement parameters	Observed measurements
$f_L = 4.25\text{GHz}$	$f_1 = 4.3\text{GHz}$	$f_{z1} = 14\text{kHz}$
$f_H = 4.35\text{GHz}$	$f_2 = 4.3\text{GHz}$	$f_{z2} = -13\text{kHz}$
$\rightarrow f_H - f_L = 100\text{MHz}$	$T_o = 50\mu\text{s}$	
$T_s = 1\text{ms}$		

$$\begin{aligned} v &= \frac{3 \cdot 10^8 \text{m/s}}{2} \cdot \frac{1 \cdot 10^3 \text{Hz}}{8.6 \cdot 10^9 \text{Hz}} = 17.65 \text{m/s} \\ d_o &= \frac{\left(3 \cdot \frac{10^8 \text{m}}{\text{s}}\right) (1 \cdot 10^{-3} \text{s})}{2(100 \cdot 10^6 \text{Hz})} \cdot \frac{27 \cdot 10^3 \text{Hz}}{2} = 20.24 \text{m} \end{aligned}$$

Do you see any problem(s) with this system?

FMCW Radar – Range/Velocity Measurement



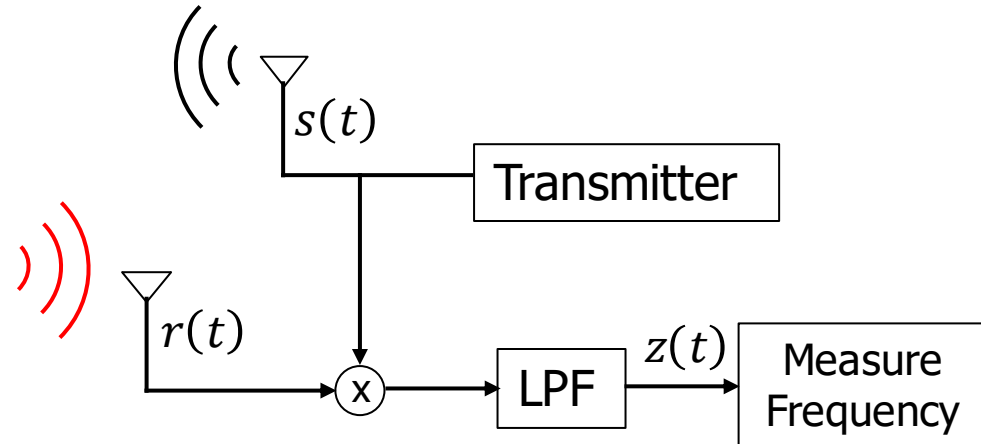
Problem #1: Our structure for measuring frequency difference does not provide us with the sign, only the absolute value of the frequency difference.

If $s(t) = A \cos(\omega_1 t)$,
and $r(t) = B \cos(\omega_2 t)$,
then $z(t) = \frac{AB}{2} \cos((\omega_1 - \omega_2)t)$.

Note that since cosine is an even function then $\cos((\omega_1 - \omega_2)t) = \cos((\omega_2 - \omega_1)t)$.

Therefore, this device cannot tell whether the frequency of $r(t)$ is bigger or smaller than the frequency of $s(t)$.

It is up to us to infer which signal has the larger frequency.



What is the solution to Problem #1?

Note: The frequency difference depends on the various system parameters according to

$$f_z \cong \frac{2vf_1}{c} \pm \underbrace{\frac{2d_o}{c} \cdot \frac{f_H - f_L}{T_s}}_{\text{This term is always positive.}}$$

This term is always positive.

This term tends to be smaller in magnitude than the other.

FMCW Radar – Range/Velocity Measurement



Solution to Problem #1: Since we are only able to observe $|f_{z_1}|$ and $|f_{z_2}|$, solve the system of equations 4 times, once for each possible combination of signs that might be associated with the two measurements. Only one of the solutions will make sense (hopefully).

For the previous example, we observed: $|f_{z_1}| = 14\text{kHz}$, $|f_{z_2}| = 13\text{kHz}$

Option 1: $f_{z_1} = 14\text{kHz}$, $f_{z_2} = 13\text{kHz} \rightarrow v = 476.5\text{m/s}$, $d_o = 0.5\text{m}$

Option 2: $f_{z_1} = 14\text{kHz}$, $f_{z_2} = -13\text{kHz} \rightarrow v = 17.5\text{m/s}$, $d_o = 20.2\text{m}$

Option 3: $f_{z_1} = -14\text{kHz}$, $f_{z_2} = -13\text{kHz} \rightarrow v = -476.5\text{m/s}$, $d_o = -0.5\text{m}$

Option 4: $f_{z_1} = -14\text{kHz}$, $f_{z_2} = 13\text{kHz} \rightarrow v = -17.5\text{m/s}$, $d_o = -20.2\text{m}$

- ✓ The last two options are eliminated because range to the target cannot be negative.
- ✓ The first option is eliminated as the velocity/range of the target is out of the expected range of reasonable values

FMCW Radar – Range/Velocity Measurement



Problem #2: In the previous example, our observation window was not long enough to provide sufficient frequency resolution.

$$T_o = 50\mu s \rightarrow \text{Frequency resolution} \sim \frac{1}{50\mu s} = 20kHz$$

Solution to Problem #2: Make T_o larger. We will have to make T_s larger as well.

Example: Suppose we make $T_s = 50ms$, $T_o = 5ms$.

$$T_o = 5ms \rightarrow \text{Frequency resolution} \sim \frac{1}{5ms} = 200Hz$$

- ✓ For the numbers of the previous example ($v = \frac{17.5m}{s}$, $d_o = 20m$) the above parameter changes would lead to measured frequency differences of $|f_{z_1}| = 635Hz$, $|f_{z_2}| = 357Hz$. A frequency resolution of 200Hz may lead to substantial errors in our estimated range/velocity values.
- ✓ Furthermore, increasing T_s and T_o may exacerbate problem #1.

Alternative solution to problem #2: There is a class of signal processing techniques (super-resolution algorithms) that provide better frequency resolution without the need to increase the observation interval.

Want to learn more about signals and systems?



- ✓ To understand the impact of noise on systems, you will need to have a solid understanding of probability and random processes → ECEN 303.
- ✓ To study modern digital communication systems → ECEN 455
- ✓ To study digital signal processing techniques → ECEN 444
- ✓ To study control systems → ECEN 420
- ✓ Pattern Recognition → ECEN 429
- ✓ Digital Image Processing → ECEN 447